Module 6_Assignment_Andrew Claxton_QMM

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```
library(lpSolve)
obj <- c(
  622, 614, 630, 600, # Plant A
  641, 645, 649, 625  # Plant B
)
```

Load library lpSolve. Establish objective vector where the variables represent combined production and shipping costs from plant A and Plant B to Warehouses 1, 2,3 plus a dummy warehouse for both plants to account for unused capacity.

```
const.mat <- matrix(c(
  # Plant A
  1,1,1,1,0,0,0,0,

  # Plant B
  0,0,0,0,1,1,1,1,

  # Wh1
  1,0,0,0,1,0,0,0,

  # Wh2
  0,1,0,0,0,1,0,0,

  # Wh3
  0,0,1,0,0,0,1,0,

  # Dummy whs
  0,0,0,1,0,0,0,1
), nrow = 6, byrow = TRUE)
```

Establish the constraint matrix with rows for the plant A capacity constraint, the plant B capacity constraint, wharehouse 1 demand, warehouse 2 demand, warehouse 3 demand, and dummy destinations

```
const.dir <- c(
  "<=", # Plant A
  "<=", # Plant B
  "=",  # W1
  "=",  # W2
  "=",  # W3
  "="   # Dummy
  )
```

Establish constraint directions. Plants can produce less than or equal to their capacity but warehouse requirements must be met exactly to meet demand. The dummy contraint direction is also set to equal to account for the entire 10-unit surplus.

```
rhs <- c(
  100,
  120,
  80,
  60,
  70,
  10
)
```

Establish right hand side constraint values for plant production capacity, warehouse demand, and the 10-unit surplus capacity.

```
solution <- lp(
  direction = "min",
  objective.in = obj,
  const.mat = const.mat,
  const.dir = const.dir,
  const.rhs = rhs
)
```

Call the lp function. Set direction to minimize, then use our variables above the objective vectors, constraint matrix, constraint directions, and rhs values.

```
matrix(
solution$solution,
nrow = 2,
byrow = TRUE,
dimnames = list(
c("PlantA","PlantB"),
c("W1","W2","W3","Dummy")
)
)
```

```
##        W1 W2 W3 Dummy
## PlantA  0 60 30    10
## PlantB 80  0 40     0
```

Our result satisfies all constraints. Plant A will produce and ship 90 units, shipping 60 to warehouse 2 and 30 units to warehouse 3. Plant B will produce 120 units, shipping 80 to warehouse 1 and 40 to warehouse 3. The solution accounts for 10 extra production capacity and assigns it to plant A because that plant has the lower production cost, but a real company likely wouldn't actually produce those units unless they could be effectively used in the next cycle of shipping. This solution also logically makes sense based on plant A's high shipping costs to warehouse 1 and plant B's high shipping costs to warehouse 2.